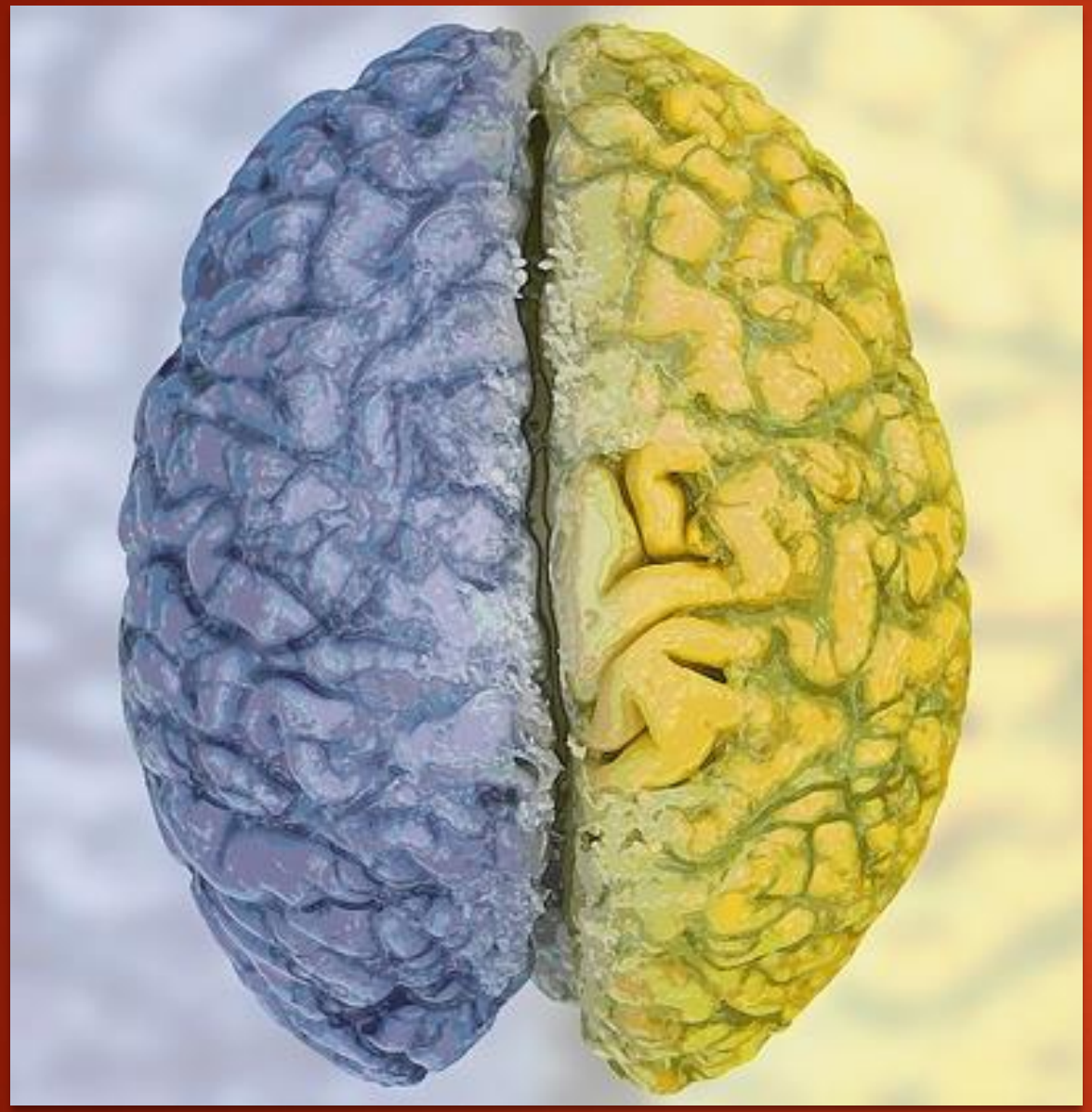


Loyola Neuroscience: Registration Presentation and Q&A

NEUROSCIENCE SOCIETY MEETING & MENTORING PROGRAM EVENT

NOVEMBER 5, 2026

WWW.LUC.EDU/NEUROSCIENCE

Neuroscience Co-Directors

Bill Rochlin

(Cellular/Molecular)



wrochli@luc.edu

Eric Gobel

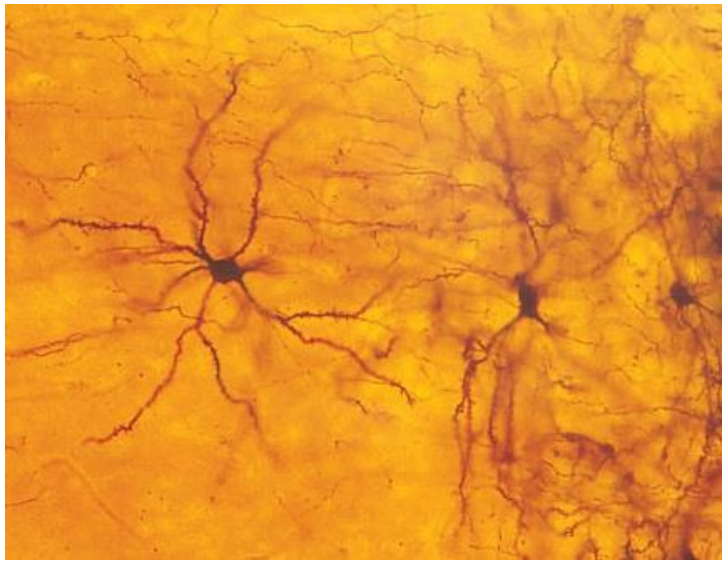
(Cognitive/Behavioral)



egobel@luc.edu

Neuroscience Majors and Minor

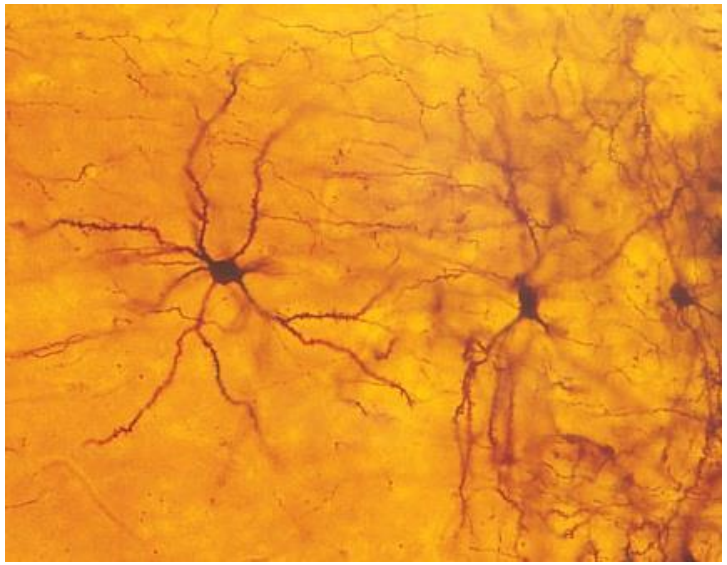
- **Molecular/Cellular Neuroscience Major (NRMC-BS)**
  - Studying neuronal structure/function, biochemical/genetic underpinnings of behavior, nervous system development
  - Neurobiological techniques in animal models or cell culture; computational techniques
- **Cognitive/Behavioral Neuroscience Major (NRCB-BS)**
  - Studying the neural processes underlying perception, behavior, and cognition (e.g., memory, language, emotions, thinking, decision-making, consciousness…)
  - Pairs psychological tasks with neuroscientific techniques in humans or animal models; computational techniques
- **Both tracks**: Understanding how the brain works, neural diseases and treatments, excellent prep for medical/graduate school



- **Neuroscience Minor (NRSC-MINR)**
  - Fundamentals of neuroscience
  - Explore the field with less extensive course requirements

Neuroscience Major Career Paths

- **Graduates from both tracks of the Neuroscience Major are well-equipped to pursue careers in many fields, including:**
  - Medicine, Dentistry, PA, Nursing (after professional school)
  - Neuroscience-related research (after graduate school)
  - Pharmaceutical and neuro-/biotechnology industry
    - Develop therapeutics or prosthetics for the nervous system
  - Neuro-/bioengineering
  - Allied health professions
    - Optometry, physical therapy, acupuncture, genetic counseling, CT technician, clinical psychology, etc.
  - Data analytics or data science
  - Science education and outreach
  - Scientific/technical writing
  - NeuroX careers (X = law, ethics, marketing, design, etc.)



Neuroscience Curriculum

Typical Class Sizes

Most Lectures are 35-48

Labs are 12-24

Capstone Seminars are 20-24

Consult Loyola Catalog for full course list

Required of all Majors

Gen Bio + Lab (8)

First-Year Chem + Lab (8)

Physics + Lab (8)

Calculus + Statistics (6)

Intro to Psych (3)

Cell Bio + Genetics (6)

Cell Bio OR Genetics Lab (1)

Neuroscience Core

Intro to Neuroscience (3)

Neurobiology (3)

Behav/Cog Neurosci (3)

Molecular/Cellular

Second-Year Chemistry (6)

3 lecture electives (7-9)

2 lab electives (3-8)

NRMC Capstone Seminar (3) or Developmental Neurobiol (3)

Cognitive/Behavioral

Research Methods (3)

Computer Science course (3)

Neuroscience Seminar (1)

3 lecture electives (9)

2 lab electives (5-7)

NRCB Capstone Seminar (3)

Neuroscience Minor

Intro to Neuroscience (3)

Neurobiology (3)

Behav/Cog Neurosci (3)

Neuroscience Seminar (1)

2 lab electives (4-8)

1 lecture or lab elective (2-4)

Suggested First-Year Sequence

Fall Semester

BIOL 101	General Biology I	3
BIOL 111	General Biology I Lab	1
CHEM 160	Chemical Structure and Properties	3
CHEM 161	Chemical Structure and Properties Lab	1
PSYC 101	General Psychology	3
MATH 131	Applied Calculus I	3
UCWR 110	Writing Responsibly	3
UNIV 101	First Year Seminar	1
Total Credits		**18**

Spring Semester

BIOL 102	General Biology II	3
BIOL 112	General Biology II Lab	1
CHEM 180	Chemical Reactivity I	3
CHEM 181	Chemical Reactivity I Lab	1
NEUR 101	Introduction to Neuroscience	3
CORE	Core Class	3
CORE	Core Class	3
Total Credits		**17**

Neuroscience Major & Other Programs

- **Double-majoring with PSYC or BIOL**; or combining with PSYC or BIOL minors
  - Unlimited double-dipping with these majors and minors makes these combinations rather feasible

- Other double-majors are difficult, but combining with other minors is doable
  - Including Bioethics or Data Science

- Either track is an **excellent major for pre-medical/pre-health students**
  - Additional courses needed not already **required** by major:
    - **NRMC**: Biochemistry (CHEM 361/BIOL 366 [NRMC lecture]), Sociology (SOCL 101)
    - **NRCB**: Second-Year Chem (CHEM 240 & 260), Biochemistry (CHEM 361), SOCL 101
    - May be recommended: Human Structure and Function I & II (BIOL 242 & 243) and Microbiology (BIOL 302)

Scheduling Considerations

- Take care of required courses ASAP (okay to delay Physics if needed)
  - Many lectures and labs in BIOL require **Cell Bio (BIOL 251)** and/or **Genetics (BIOL 282)**; some require Second-Year Chemistry (CHEM 240 and 260)
  - Most labs in PSYC require **Stats (PSYC 304)** and **Methods (PSYC 306)**
- NRCB majors should take NEUR 300 (Neuroscience Seminar) ***before*** NEUR 387 (Seminar in Behavioral and Cognitive Neurosciences)
- Engaged Learning (EL) courses
  - Options within the major are **BIOL 373 Lab in Neurophysiology and Behavior**, **CURE labs**, or an appropriate **Independent Research** project (conducting neuroscience major track-relevant research in BIOL 396, PSYC 397, or equiv.)
  - **Not required** to take EL course in the major (e.g., could take EL course in SES)

Spring 2027 Scheduling Notes

- **Biology Special Topics Courses**
  - **BIOL 395-002 Glial Biology in Health and Disease** (pre-reqs: NEUR 101, BIOL 251, BIOL 282)
    - Counts as **Specialty Lecture** in **NRMC Major and NRSC Minor**
  - **BIOL 395L 01E: Cancer Cell Culture** (pre-req: BIOL 251)
    - Counts as **Specialty Lab** in **NRMC Major and NRSC Minor** if project involves **glioblastoma cells**
  - **BIOL 395L 02E CURE: Epigenetics Research** (pre-req: BIOL 282)
    - Counts as **Specialty Lab** in **NRMC Major**
  - May need to have your CAS advisor manually enter these courses into your ARR
- **CHEM 366L Biochemistry Lab (NRMC Lab)** – LOCUS registration (mostly) restricted to Forensic Science students in Sp27 (cannot register on LOCUS)
  - Contact the course instructor Jessica Eisenberg (jeisenberg2@luc.edu) to get on the waitlist
- **COMP 180 Computing and Data Analysis for the Sciences – NRCB Comp Sci requirement**
  - **Mon 4:15-6:45pm**: Created primarily to **support students in Neuroscience and Psychology** wanting an **introduction to computing focused on working with data** (MATLAB-based)

Spring 2027 Newer Courses

- **Computational Neuroscience Sequence**
  - **Fall: PSYC 383/COMP 386 Methods in Computational Neuroscience**
    - Pre-reqs: NEUR 101; MATH 131/161; COMP 150/170/180 or DSCI 101)
    - **Neuroscience Specialty Lecture**; Psychology and Computer Science Elective
  - **Spring: PSYC 384/COMP 389 Computational Modeling of the Brain** (pre-req: PSYC 383/COMP 386)
    - **Neuroscience Specialty Lab**; Psychology and Computer Science Elective
- **Cellular and Gross Neuroanatomy Courses**
  - **BIOL 361 Cellular and Gross Neuroanatomy Lecture - Specialty Lecture**
    - Pre-reqs: NEUR 101 and BIOL 251 (can take **with or without** Neuroanatomy Lab)
  - **BIOL 361L Cellular and Gross Neuroanatomy Lab - Specialty Lab**
    - **Pre-req or co-req**: BIOL 361 Neuroanatomy Lecture (**current or previous semester**)

Spring 2027 Available Specialty Lectures/Labs

Specialty Lectures	NRCB	NRMC	NRSC
BIOL 307 Biology of Stem Cells		X	
BIOL 357 Neural Disease, Degeneration, Regeneration (2 sections)	X	X	X
BIOL 361 Cellular and Gross Neuroanatomy	X	X	X
BIOL 382 Molecular Genetics		X	
BIOL 395-002 Glial Biology in Health & Disease		X	X
CHEM 361 Biochemistry or CHEM 370 Biochemistry I		X	
PSYC 240 Sensation & Perception	X	X	X
PSYC 350 Developmental Cognitive Neuroscience	X		X
PSYC 378 Drugs and Behavior	X	X	X

Specialty Lab	NRCB	NRMC	NRSC
BIOL 361L Cellular and Gross Neuroanatomy Lab	X	X	X
BIOL 366L Biochemistry Lab*		X	X
BIOL 373 Lab in Neurophysiology and Behavior	X	X	X
BIOL 390 Molecular Biology Lab		X	X
BIOL 395L-01E Cancer Cell Culture (glioblastoma project)		X	X
BIOL 395L-02E CURE Epigenetics Research		X	
PSYC 312 Lab in Cognitive Neuroscience	X		X
PSYC 314 Lab in Cognitive Psychology (3 sections)	X		X
PSYC 316 Lab in Sensation & Perception (2 sections)	X	X	X
PSYC 384/COMP 389 Compu-tational Modeling in the Brain	X	X	X

J-Term 2027 Courses (all online)

- BIOL 111 General Biology Lab I
- BIOL 251 Cell Biology
- BIOL 282 Genetics
- BIOL 395L Special Topics Laboratory (Computational Neurosciecne Lab) taught by Dr. Hui Ye
  - Prereq: NEUR 101 and BIOL 251, or by permission. Students are preferred to have taken college level course in algebra and general physics. Computer coding skill is preferred but not required.
  - Counts as **Specialty Lab** for **NRCB and NRMC majors, NRSC minor**
    - May need to have your CAS advisor manually enter this course into your ARR
- COMP 150 Introduction to Computing
- PSYC 304 Statistics

Summer 2027 Preliminary Course Offerings (most are online)

- Most Science/Math Foundations, including Psychology Stats/Methods and Computer Science requirements

- Neuro Core: **NEUR 101 Introduction to Neuroscience**

- Neuroscience Specialty Labs
  - **PSYC 314 Lab in Cognitive Psychology** (NRCB major, NRSC minor, PSYC major/minor)
  - **BIOL 373 Laboratory in Neurophysiology and Behavior** (in-person EL course)
  - **BIOL 398 Internship in Biology** (EL course requiring outside research lab placement; may count as Specialty Lab if research is related to major)

Undergraduate Research

- **Many of our students conduct neuroscience-related research in faculty research labs at Loyola or other Chicago-area research institutions**
  - Conducting research with a neuroscience track-related focus can count as a **Neuroscience Specialty Lab** and **EL-Independent Research credit**
    - Must complete an **EL-Independent Research course** (**BIOL 396 or PSYC 397**, or another department/institution with special arrangement; requires writing an empirical report and/or presentation of your research study)
    - Contact your major track's director for approval process
  - Should start working in a lab earlier rather than later (may need 1-2 semesters in a lab before taking EL-Independent Research)
- **Honors in Neuroscience**
  - **NRMC majors**: Honors with NRMC GPA > 3.8 (overall GPA > 3.5); High Honors if also earn an A in independent research/internship course with neuroscience focus
  - **NRCB majors**: Honors with overall GPA > 3.5 and completion of neuroscience-focused honors thesis research in PSYC 369/370 sequence with B or higher
- **Pizzi Faculty Mentors Fellowship Program (academic year LUROP)**
  - Two Neuroscience majors annually are awarded $8k fellowship (up to $3.5k stipend and $1k travel) to support year-long neuroscience research project in lab(s) at Loyola

Undergrad Research with Neuro Faculty

NTT Faculty:
Eric Gobel (PSYC)
Jason Pych (PSYC)
Stephen Logan (BIOL)



Nicholas Baker
shape/object perception



Dan Cavanaugh
circadian system



Yanan Chen
inflammatory demyelination



Jary Delgado
synaptic plasticity



Norberto Grzywacz
neuroaesthetics



Maggie Guy
cognitive development



Bob Morrison
cognitive neuroscience



Bill Rochlin
developmental neurobiology



Becky Silton
affective neuroscience



Stephan Steidl
motivated behavior



Stephanie Grella
learning and memory updating



Liz Wakefield
gesture and action



Hui Ye
neurodegenerative diseases



Wei-Ming Yu
auditory development

Student Support and Opportunities

- **Tutors for NEUR 101** and supplemental instruction/tutoring for many foundational science courses

- **Studying Abroad as a Neuroscience Major**
  - **DIS in Scandinavia (Copenhagen, Denmark or Stockholm, Sweden)**: Excellent new program with many neuroscience specialty lectures and labs available!
  - **Rome Campus**: Only courses in Neuroscience curriculum offered are Cell Biology (fall) and Genetics (spring), so requires careful planning or must go in summer session
  - Other study abroad programs require careful planning or summer/embedded program

- **Accelerated Masters Program (AMP) in Biology**

- **Other Neuroscience Programs in Development**
  - **Accelerated MS in Neuroscience** degree for advanced undergraduates who have begun neuroscience research or internship by junior year, including biotech industry track
  - **Socio-Environmental Neuroscience** undergraduate major for students interested in interactions between societal/environmental factors and nervous system
  - **Computational Neuroscience** undergraduate minor

Neuroscience Seminar (NEUR 300)

- Loyola Neuroscience faculty and invited outside speakers from Chicago area and beyond; weekly during fall and spring semesters
  - Presenting about their research or professional development topics
  - Learn about and meet faculty with whom to conduct research!
- Open to all; encouraged to attend even if not enrolled in the course
- **Tuesdays at 4:00 PM in Cuneo 109** (currently also a Zoom option)
- Link to schedule: www.luc.edu/neuroscience/seminar

Student Organizations

- **Neuroscience Society** (advisor: Dr. Morrison)
  - Bi-weekly meetings
  - Visiting speakers, social events, outreach, fundraising (e.g., Walk to End Alzheimer's)
  - Coordinates **Loyola Neuroscience Mentoring Program (LNMP)**
  - Hosts **Annual Neuroscience Banquet** in the spring (Invited speaker, student awards, N$\rho\Psi$ induction, dinner)
- **Loyola chapter of the N$\rho\Psi$ (Nu Rho Psi) Honor Society** (advisor: Dr. Wakefield)
  - Membership by invitation to neuroscience majors or minors who have met the chapter's minimum neuroscience coursework and GPA criteria
- **Loyola Women in Neuroscience (LWiN) Association** (advisor: Dr. Grella)
- **Brain Exercise Initiative (BEI)** (advisor: Dr. Wakefield)
- **Loyola Learning and Memory Association (LLaMA)** (advisor: Dr. Grella)
- **Neuro & Nature Alliance LUC**